Department of Chemical Engineering, Chulalongkorn University

Teaching document · operator training simulator

Cyclohexane Reactor Ver 1

CAUSE AND EFFECT MATRIX

Safety instrumented system, alarms and start permissives

Document no.	**CHX-V1-IN-CE-001**
Unit	Cyclohexane Reactor Ver 1
Client	Department of Chemical Engineering, Chulalongkorn University
Title	Cause and effect matrix · safety instrumented system, alarms and permissives
Drawings	CHX-V1-PID-001 (screen P&ID, P&ID-001) · PFD CHX-V1-PFD-001
Revision	R0 · 2026-10-26 · Issued for teaching
Sheets	Cover + 1 sheet A3 landscape

45 cause rows: 7 unit trips incl. the ESD pushbutton, 28 alarm limits, 10 start permissives and interlocks · 12 effect columns

Every setting, delay and effect on this sheet was read from the simulator model (model.js: K, TRIPDLY, tripConditions(), bmsTrip(), bmsStep(), bmsCmd(), purgePermissives(); ui1.js: the alarm table ALM) by tabs/cem/tools/sf_data.py and proven by running the model; the C&E matrix tab of the simulator checks them again against the running page. This is a teaching document: it is not the C&E of a real plant, and checker and approver are left blank on purpose.

REVISIONS

REV	DATE	DESCRIPTION	ORIGINATOR	CHECKER	APPROVER
R0	2026-10-26	Issued for teaching	Apinan Soottitantawat		

CAUSE AND EFFECT MATRIX · CYCLOHEXANE REACTOR VER 1

Cause and effect matrix · safety instrumented system, alarms and permissives

CAUSES								EFFECTS											
ITEM	TAG	SERVICE	DIR.	SETTING	UNIT	DELAY s	NOTE	**XV-101** Benzene feed SIS valve	**XV-105** Make-up H2 SIS valve	**K-101** Recycle compressor	**FIC-101** Benzene feed FV-101	**FIC-103** H2 to sparger FV-103	**PIC-105** Make-up H2 PV-105	**HS-101** E-101 COOL, HV-110 shut	**FV-104** Pump-around to E-101	**HS-102** R-102 jacket, TV-102 100 %	**EDV-101** Depressuring to flare	**1ST-OUT** First-out on SIS panel	**ALARM** DCS alarm and horn
							TRIP ACTION	C	C	STOP	MAN 0 %	MAN 0 %	MAN 0 %	COOL	FO 100 %	COOL	O	FO	A
001	**HS-100**	Emergency shutdown pushbutton (operator)	ON	manual		—	Latched; the only cause that opens EDV-101	X	X	X	X	X	X	X	X	X	X	X	A
002	**TSHH-100**	R-101 temperature high-high	HH	220	°C	5	Element TE-100, not the TIC-101 element; latched	X	X	X	X	X	X	X	X	X		X	A
003	**TSHH-107**	R-103 bed temperature high-high	HH	220	°C	5	Hottest of TI-107A…E; latched	X	X	X	X	X	X	X	X	X		X	A
004	**PSHH-101**	R-101 pressure high-high	HH	33.5	bar g	3	PT-101 at R-101 (PIC-105 is at V-104); latched	X	X	X	X	X	X	X	X	X		X	A
005	**XS-101**	Agitator M-101 stopped with XV-101 open	STOP	stopped		10	Run signal; armed only with XV-101 open; latched	X	X	X	X	X	X	X	X	X		X	A
006	**XS-102**	Compressor K-101 stopped with XV-101 open	STOP	stopped		10	Run signal; armed only with XV-101 open; latched	X	X	X	X	X	X	X	X	X		X	A
007	**LSHH-104**	V-104 level high-high: liquid toward K-101	HH	90	%	5	LT-104S; not latched, no first-out; XS-102 trips 10 s later if XV-101 open			X									A
008	**FAL-101**	Benzene feed low	L	25	t/h	—	Suppressed while no benzene flows												A
009	**FAH-101**	Benzene feed high	H	42	t/h	—	Suppressed while no benzene flows												A
010	**FAL-103**	Hydrogen to R-101 sparger low	L	35 000	Nm³/h	—	Suppressed while no benzene flows												A
011	**PAL-105**	Loop pressure (V-104) low	L	27.5	bar g	—	Suppressed without H2 in the loop												A
012	**PAH-105**	Loop pressure (V-104) high	H	30.5	bar g	—	Suppressed without H2 in the loop												A
013	**TAL-101**	R-101 temperature low	L	190	°C	—	Suppressed while no benzene flows												A
014	**TAH-101**	R-101 temperature high	H	208	°C	—	Suppressed while no benzene flows												A
015	**TAH-100**	R-101 temperature (independent element) high	H	210	°C	—	Always armed												A
016	**LAL-101**	R-101 level low	L	45	%	—	Suppressed while no benzene flows												A
017	**LAH-101**	R-101 level high	H	82	%	—	Suppressed while no benzene flows												A
018	**JAL-101**	Agitator M-101 power low	L	80	kW	—	Suppressed while M-101 is stopped												A
019	**TAL-102**	R-102 temperature low	L	148	°C	—	Suppressed while no benzene flows												A
020	**TAH-102**	R-102 temperature high	H	172	°C	—	Suppressed while no benzene flows												A
021	**LAL-102**	R-102 level low	L	30	%	—	Suppressed while no benzene flows												A
022	**LAH-102**	R-102 level high	H	80	%	—	Suppressed while no benzene flows												A
023	**PDAH-102**	Catalyst filter F-102 ΔP high	H	0.8	bar	—	Always armed												A
024	**TAH-107**	R-103 hottest bed temperature high	H	185	°C	—	Always armed; hottest of five cells												A
025	**LAL-104**	V-104 separator level low	L	25	%	—	Suppressed while no benzene flows												A
026	**LAH-104**	V-104 separator level high	H	78	%	—	Suppressed while no benzene flows												A
027	**AAH-103**	Product benzene (spec 10 ppm) high	H	10	ppm	—	Suppressed while no benzene flows												A
028	**AAH-104**	Product MCP isomer (spec 200 ppm) high	H	200	ppm	—	Suppressed while no benzene flows												A
029	**AAL-105**	Recycle hydrogen purity low	L	85	% H2	—	Suppressed without H2 in the loop												A
030	**LAL-109**	D-101 steam drum level low	L	25	%	—	Always armed												A
031	**LAH-109**	D-101 steam drum level high	H	75	%	—	Always armed												A
032	**PAL-110**	D-101 steam pressure low	L	3.8	bar g	—	Suppressed unless E-101 is on COOL												A
033	**PAH-110**	D-101 steam pressure high	H	5.6	bar g	—	Suppressed unless E-101 is on COOL												A
034	**TAH-103**	E-103 vent temperature (reflux condenser) high	H	75	°C	—	Suppressed while no benzene flows												A
035	**AAH-100**	Loop oxygen with hydrogen admitted high	H	1	% O2	—	Suppressed without H2 in the loop												A
036	**SIS-100**	SIS trip latched (not reset)	TRIP	latched		—	Reset refused while a trip is active or TI-100 > 210 °C	P	P	P									
037	**AI-100**	Loop oxygen high before hydrogen	H	1	%	—	Purge with N2 first; H2 and benzene permissive	P	P										
038	**AI-105**	H2 not admitted or loop purity low	L	80	%	—	Needs XV-105 open	P											
039	**LIC-101**	R-101 level low (agitator, sparger covered)	L	40	%	—	Benzene start permissive	P											
040	**M-101**	Agitator not running	STOP	stopped		—	Benzene start permissive	P											
041	**FIC-103**	K-101 off or H2 flow to sparger low	L	15 000	Nm³/h	—	Benzene start permissive	P											
042	**TI-100**	R-101 below light-off temperature	L	140	°C	—	Benzene start permissive	P											
043	**HV-113**	E-103 reflux condenser CW not open	L	50	% open	—	Benzene start permissive	P											
044	**PIC-105**	K-101 suction pressure low	L	10	bar g	—	K-101 start interlock (pressure the loop first)			P									
045	**LT-104S**	V-104 level high, liquid toward K-101	H	85	%	—	K-101 start interlock			P									
046	SPARE																		
047	SPARE																		

Numbers from the simulator model (sf_data.py)

NOTES

1. X = action on the cause; P = permissive: the valve cannot be opened (or the motor started) while the cause is present; A = DCS alarm and horn only.
2. Every unit trip is latched: reset on the SIS tab; the reset is refused while a non-motor trip condition is active or TI-100 is above 210 °C.
3. LSHH-104 stops K-101 only (not latched, no first-out). With XV-101 open, XS-102 trips the unit 10 s later (model: K-101 stopped at 5.25 s, unit trip at 15.5 s).
4. XS-101 and XS-102 are armed only while XV-101 is open (benzene permitted).
5. While the SIS is TRIPPED the model forces FV-104 and TV-102 to 100 % on every step, whatever their controller asks.
6. Not done by a trip: P-100A, P-101A, P-102A, M-101 keep running; LIC-101, LIC-102, LIC-104, LIC-109 stay in AUTO; the purge HIC-107 is unchanged.
7. Delay = time the cause must persist before the action; the model acts on the next 0.25 s step after the delay.
8. Alarm rows have no delay; suppression follows the DCS console (for example process alarms are suppressed while no benzene flows).
9. Only the ESD pushbutton HS-100 opens EDV-101 (depressuring to flare); EDV-101 can also be opened by hand on the SIS tab.
10. ALM also lists a high limit of 56 000 Nm³/h for FIC-103; the DCS never raises it (evalAlarms arms a high alarm only when the limit is below 900), so it is not a row here.
11. Settings, delays and effects come from the simulator model (K, TRIPDLY, ALM, tripConditions(), bmsTrip(), bmsStep()); the C&E tab re-checks them against the running page.

LEGEND

X	action on the cause (trip action row says which)
P	permissive: cannot be opened / started while the cause is present
A	DCS alarm and horn only
TAG	red tag = unit trip (SIS); black = alarm or permissive

Cyclohexane Reactor Ver 1 · operator training simulator (benzene hydrogenation, 300 kt/y)	
Client	Department of Chemical Engineering, Chulalongkorn University
Project	Cyclohexane Reactor Ver 1 · OTS (teaching)
Document no.	**CHX-V1-IN-CE-001**
Title	Cause and effect matrix · SIS, alarms and permissives
P&ID	CHX-V1-PID-001 (screen P&ID, P&ID-001)
Status	Rev 0 · Issued for teaching · sheet 1 of 1

REV	DATE	DESCRIPTION	ORIGINATOR	CHECKER	APPROVER
R0	2026-10-26	Issued for teaching	Apinan Soottitantawat		